



Silica-based nanomaterials for the control of *Trioza anceps*: Laboratory bioassays and preliminary field observations

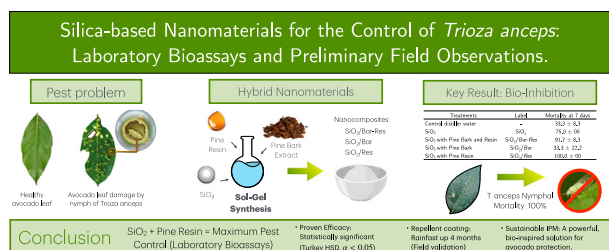
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Received: 5 January 2026 / Accepted: 16 April 2026 / Published online: 26 May 2026
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Abstract

It is known that four *Trioza* species attack avocado foliage, with *Trioza anceps* Tuthill (commonly called the avocado psyllid) being a widespread pest in Mexico and Central America. Both nymphs and adults feed on young leaf sap, but the nymphs induce avocado leaf galls by injecting a toxin during feeding, triggering tissue hyperplasia and hypertrophy. This leads to severe leaf deformations, stunted tree growth, and overall decline in plant health [1]. The aim of the present research was to evaluate the efficacy of nanomaterials synthesized via the sol-gel method for the control of *T. anceps*. The treatments included silicon oxide, silicon oxide with pine bark extract, silicon oxide with pine resin, and silicon oxide with both pine bark extract and pine resin. The nanomaterials were characterized prior to laboratory testing. Among them, the SiO₂-pine resin formulation proved to be the most effective, achieving 100% mortality of *T. anceps* nymphs by day 7 and belonging to the highest statistical efficacy group (group a) according to Tukey's HSD test ($p < 0.05$). in [2]. Material characterization confirmed physicochemical properties enhancing phytoprotective effects, positioning silica-based nanomaterials as sustainable alternatives in modern agriculture [3, 4].

Graphical Abstract



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Keywords Avocado psyllid gall · Pest control · Silica (SiO₂) · Pinus resin

Highlights

- Sol-gel synthesized SiO₂-pine resin nanomaterial achieved 100% mortality of *T. aniceps* nymphs by day 7 ($p < 0.05$), demonstrating hybrid organic-inorganic matrix potential.
- Characterization confirmed distinct optical fingerprints (band gap 4.7–5.5 eV) and functional groups (Si-O-Si, C=O) and nanostructure morphology essential for organic-inorganic bonding.
- SiO₂/resin showed strong rainfastness (4 months leaf visibility) and caused no adverse effects on non-target entomofauna (predators, pollinators, parasitoids) confirming matrix stability under field conditions.

1 Introduction

Nanomaterials have emerged as fundamental tools in modern agriculture, bridging the gap between technological innovation and sustainable practices [5]. Their integration into agronomic systems -from precision farming to integrated pest management- has opened new frontiers in crop optimization. The unique physicochemical properties of nanomaterials, particularly their high surface area to volume ratio and quantum confinement effects, enable unprecedented control over biological interactions at the molecular level. These characteristics facilitate: (i) enhanced targeting efficiency of active compounds, (ii) improved nutrient bioavailability, and (iii) tunable release kinetics for sustained crop protection.

Particularly in phytosanitary applications, nano-enabled formulations demonstrate remarkable potential to overcome limitations of conventional agrochemicals [6]. Silicon dioxide (SiO₂)-based systems have garnered significant attention due to their photocatalytic properties and biocompatibility, especially recent advances in agricultural applications [3, 7]. When functionalized with plant-derived bioactive compounds, such as extracts, resins and barks, these nanostructures can potentially synergize inorganic and organic modes of action, offering a sustainable alternative for crop protection [8–10].

Silicon dioxide (SiO₂) is widely used in the manufacturing of various glass types and ceramic products, including earthenware and porcelain. Food-grade silicon dioxide (AA-6P) is commonly used in the food industry as an anti-caking agent. Common applications include vitamin and dietary supplements, seasonings, potato-based snacks, instant noodles, and coffee. When incorporated in regulated quantities, this mineral not only complies food safety standards but also contributes to bone formation and development [7]. Silicon dioxide nanomaterials exhibit advantageous physicochemical properties that mitigate the effects of toxic elements in plants, facilitate nutrient transport through xylem tissues [11, 12], and improve crop quality by reducing losses caused by pest damage and diseases during harvest and postharvest handling [13, 14].

Avocado (*Persea americana*) stands as one of Mexico's primary crops due to its strong domestic and international demand, ranking as the fourth most exported tropical fruit globally [15]. Annual production reaches approximately 7 million tons, of which 2.3 million tons are destined for export markets, positioning it as one of the most profitable crops in the country.

The avocado industry in Michoacán stands as an economic powerhouse, generating approximately 78,000 direct jobs and contributing significantly to Mexico's export revenues. While often underestimated in local currency figures, the sector's annual exports to the United States alone surpass \$2 billion USD, positioning Michoacán as the world's dominant avocado supplier and a vital engine for regional development [15, 16].

Avocado production can be adversely affected by a variety of pests, among which *Trioza aniceps* Tuthill (Hemiptera: Triozidae) is particularly relevant in modern pest control strategies [17].

The life cycle of *Trioza aniceps* Tuthill lasts approximately 60 days [18], allowing the insect to complete five to six generations per year. Females deposit oval, yellowish eggs on the underside of young, tender avocado leaves. Upon hatching, the first-instar nymphs are yellowish and highly mobile, actively seeking young tissue on the leaf's underside, where they settle and initiate phloem feeding. Once attached, the nymph secretes phytotoxic salivary substances that elicit an immune response in the plant, leading to abnormal cell proliferation and subsequent gall formation. The gall develops as leaf tissue proliferates around the nymph, forming a protective chamber that encloses it throughout its development (Fig. 1). Within the gall, the nymph undergoes morphological changes, becoming progressively flatter and more oval. Feeding damage becomes most evident during these stages, as galls expand and typically turn reddish or brown. During the final instar, external wing pads become clearly visible, indicating the approach of metamorphosis. Upon completing development, the mature nymph exits through the base of the gall to molt into a winged adult.

Conventional management of *T. anceps* relies primarily on systemic insecticides such as thiamethoxam, contact insecticides like malathion, and preventive aerosols containing abamectin, along with cultural practices such as pruning and burning of infested branches. However, the continuous and often excessive use of chemical insecticides presents several challenges. These include the development of resistance in nymph populations, limited residual activity, and high toxicity, which poses risks to human health and negatively affect agroecosystem balance. In light of these limitations, the objective of the present study was to evaluate the efficacy of nanomaterials synthesized via the sol-gel method, specifically silicon dioxide (SiO_2), silicon dioxide with pine bark extract (SiO_2/Bar), silicon dioxide with pine resin (SiO_2/Res),

and silicon dioxide with both pine resin and bark extract ($\text{SiO}_2/\text{Bar-Res}$) for the control of *T. anceps* under laboratory conditions.

2 Materials and methods

2.1 Pine resin collection

Pine resin (oleoresin) was collected from healthy, mature *Pinus teocote* trees exhibiting natural exudation at mechanical damage sites (broken branches) on the south-eastern foothills of La Malinche volcano ($19^\circ 13' 51''\text{N}$, $98^\circ 01' 55''\text{W}$; 3100 m a.s.l.), 12 km northeast of Francisco Javier Mina, Zitlaltepec de Trinidad Sánchez Santos, Tlaxcala, Mexico (Fig. 2). This non-invasive approach minimized tree stress while capturing ecologically relevant resin under natural conditions.

Aseptic tools—a stainless-steel Rodin Model 1003 artist's spatula and Santul-brand carbon steel wedges (5.0×9.5 cm)—standardized harvest depth (≈ 2 –3 mm) and wound geometry, yielding 50 g resin/tree. Tools were sterilized with 70% ethanol between trees to prevent contamination. Collected resin was stored at 4°C in amber glass vials until use.

2.2 Synthesis of nanomaterials via sol-gel method

Silicon dioxide (SiO_2) nanomaterials, including silicon dioxide with pine bark extract (SiO_2/Bar), silicon dioxide with pine resin (SiO_2/Res) and silicon dioxide with both pine resin and bark extract ($\text{SiO}_2/\text{Bar-Res}$), were synthesized via the sol-gel method under reflux conditions at 70°C .

Base solution: 19 mL of tetraethyl orthosilicate (TEOS; Sigma-Aldrich), 80 mL of anhydrous ethanol ($\text{C}_2\text{H}_6\text{O}$, 99.8%; Sigma-Aldrich), 30 mL of deionized water, 2.0 g of polyvinylpyrrolidone (PVP, MW 86,000; Sigma-Aldrich).

Pine bark extract (Bar): 7.2 g powdered *Pinus teocote* bark extracted in 80 mL ethanol via Soxhlet (6 cycles,

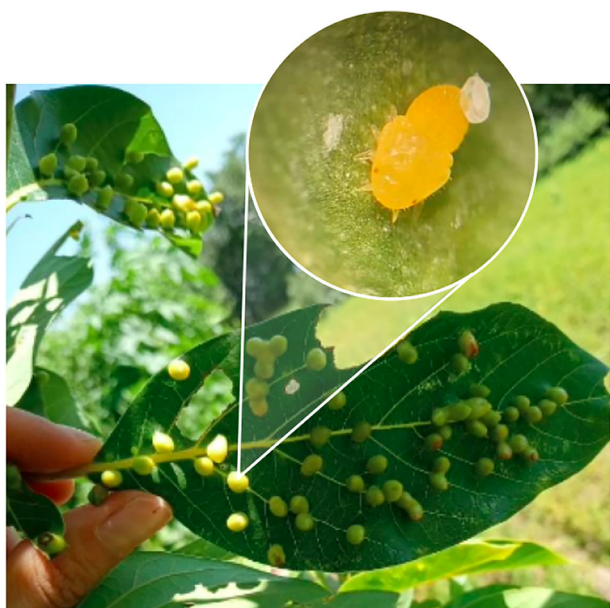


Fig. 1 Avocado leaf gall and identification of the inducer. Globular galls are visible along the avocado leaf. The inset micrograph (Leica MZ12.5, $40\times$) shows a *Trioza anceps* nymph recovered from inside a gall, the nymph is the soft-bodied insect responsible for the hypertrophic plant growth

Fig. 2 Pine resin harvesting from natural wounds: **a** Trunk showing mechanical injury resulting from branch shedding or illegal logging activities; **b** Manual collection of the solidified exudate with a stainless steel spatula

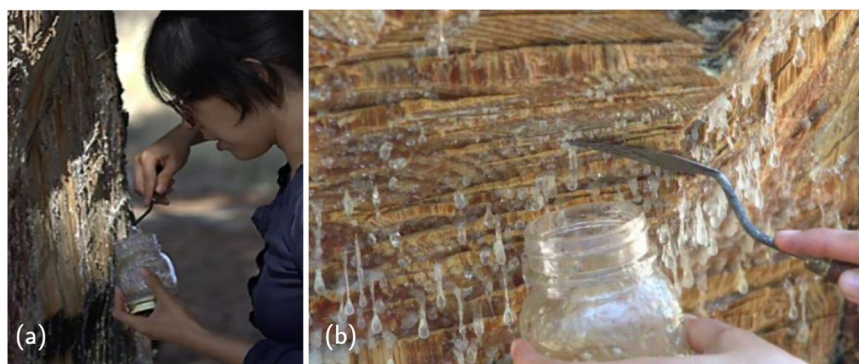


Fig. 3 Laboratory bioassays showing 16 petri dishes with leaves (four galls per leaf). Six treatments (three replicates each) were evaluated. Cotton moistened with 3 mL purified water was applied to the petiole tip to prevent wilting



heating mantle/stirring), concentrated to 20 mL by rotary evaporation (reduced pressure).

Pine resin (Res): solution: 10 g resin dissolved in 10 mL anhydrous ethanol (for SiO_2/Res and $\text{SiO}_2/\text{Bar-Res}$).

Solutions aged 24 h prior to use. Hydrolysis/condensation proceeded under stirring (4 h, 70 °C), followed by centrifugation (10,000 rpm, 15 min), washing (3 × ethanol/ H_2O), and drying (60 °C, 12 h). Final dose (1 mg/mL emulsion) justified by preliminary LD50 tests on *T. anceps* nymphs.

2.3 Nanomaterial characterization

SiO_2 , SiO_2/Bar , SiO_2/Res , and $\text{SiO}_2/\text{Bar-Res}$ materials were characterized using multiple spectroscopic and microscopic techniques following current nanotechnology protocols [19]. FTIR spectra: Digilab SCIMITAR Series spectrometer. UV-Vis spectroscopy: Varian Cary 100 spectrophotometer with integrated diffuse reflectance sphere (MgO standard, 100% reflectivity). Morphology and elemental composition: SEM-EDS with energy-dispersive spectrometer detector. FE-SEM: JEOL JSM-7800F with Oxford Instruments X-Max EDS system. Surface functional groups: Bruker Equinox 55 spectrometer with diamond ATR accessory using established sol-gel characterization methods [4, 20, 21]

2.4 Laboratory bioassays

Three completely randomized assays were conducted under controlled conditions to evaluate the mortality of *Trioza anceps* nymphs at three time intervals: 16 h, 48 h, and 7 days. A total of six treatments were tested, each with three replicates per time point. The treatments evaluated were:

Witness (distilled water).

Vehicle control (ethyl alcohol).

SiO_2 , SiO_2/Bar , SiO_2/Res , $\text{SiO}_2/\text{Bar-Res}$ (nanomaterials).

Nanomaterials were diluted 30:1 (ethanol:nanomaterial) to yield foliar-applicable emulsions (≈ 1 mg/mL; 1.3 mL/leaf, both sides; ~ 10 μL /nymph). This dose was selected based on preliminary LD₅₀ tests and field spray equivalents. The experimental unit consisted of a Petri dish containing one avocado leaf with four galls, each harboring one nymph, as shown in Fig. 1. A total of 54 leaves were used (6 treatments × 3 replicates × 3 time points). Figure 3 shows the experimental setup for a single time point (18 leaves). To maintain leaf turgor throughout the evaluation period, a cotton swab impregnated with 3 mL of purified water was placed at the distal end of the petiole. Each leaf was sprayed with approximately 1.3 mL of the corresponding treatment on both sides. Mortality was assessed post-application by stereomicroscopic examination (Leica MZ12.5, 40 × magnification) of nymphs within the galls.

The null hypothesis (H_0) states that there is no relationship between the treatments and nymph mortality; specifically, all treatments are equally effective (or ineffective). Rejection of H_0 would indicate that at least one treatment differs significantly from the others.

A Generalized Linear Model (GLM) with binomial family and logit link function was fitted to the mortality data, enabling comparison of: (i) the relative effectiveness of the six treatments within this single tree, (ii) the temporal evolution of nymph mortality across the three evaluated time points (16 h, 48 h, and 7 days), and (iii) the differences among the four nanomaterials (SiO_2 , SiO_2/Bar , SiO_2/Res , $\text{SiO}_2/\text{Bar-Res}$).

The GLM included treatment, time, and their interaction as fixed factors. Post-hoc comparisons were performed using Dunnett's test to compare each nanomaterial against the control, while Tukey's Honestly Significant Difference (HSD) test was used for multiple comparisons among the nanomaterials (SiO_2 , SiO_2/Bar , SiO_2/Res , $\text{SiO}_2/\text{Bar-Res}$).

This approach allowed the identification of significant differences between treatments at each time point and the selection of promising candidates for subsequent field validation.

All 54 leaves used in this bioassay were collected from a single avocado tree and selected for having exactly four galls each. Consequently, the results should be interpreted as a comparative evaluation of treatments under controlled conditions, not directly extrapolatable to the population level without further studies involving multiple trees and field replicates. This laboratory bioassay served as a screening tool to identify treatments showing significantly different mortality rates, warranting further evaluation. Treatments selected based on performance across all time points will be candidates for subsequent field trials with larger scale and greater representativeness to validate efficacy under natural conditions.

3 Results and discussion

3.1 Characterization of nanomaterials

The UV-Vis absorption spectra of the hydroalcoholic extract of pine bark (90% ethanol) and of pure pine resin were recorded to identify their characteristic bands and understand their contribution to the optical behavior of nanomaterials formed with these organic precursors (Fig.

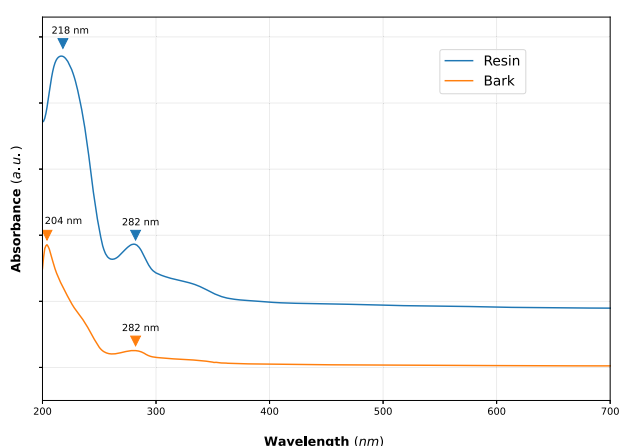


Fig. 4 UV-Vis spectra of pine bark and resin

4). The bark extract was prepared by Soxhlet extraction of 7.2 g of dried pine bark powder using 90% ethanol as the solvent, coupled to a heating mantle with integrated stirring at 70 °C, yielding 20 mL of concentrated extract. For UV-Vis analysis, the extract was diluted with 80 mL of anhydrous ethanol (C₂H₆O, 99.8%; Sigma-Aldrich). The pure pine resin was dissolved directly in anhydrous ethanol in a 1:1 weight ratio. Measurements were performed in a wavelength range of 200 to 700 nm.

The UV-vis spectra revealed both shared and distinctive absorption features, with marked differences in intensity (Fig. 4, Table 1). A common absorption band was observed at 282 nm for both spectra; however, this band was the least intense. The most prominent absorptions, with the highest amplitudes, were recorded in the far-UV region: the bark extract showed its maximum intensity at 204 nm, while the pure resin showed it at 218 nm.

The common absorption band observed at 282 nm in both spectra can be associated with shared components, mainly volatile terpenes (such as α -pinene, Δ^3 -carene, camphene, phellandrene, and p-cymene) and resin acids of the abietane and pimarane types, present in both resin and bark. However, while these compounds are dominant in the resin, in the bark the signal at 282 nm may also receive contributions from tannins and other phenolic compounds, characteristic of this tissue.

Conversely, the distinct absorption bands in the far-UV region (204 nm for bark) are attributed to their unique structural compositions. The bark's absorption at 204 nm is likely due to its high content of tannins and structural polymers like suberin and lignin, which are absent in the resin. The resin's peak at 211 nm can be assigned to its high concentration of specific terpenoids and organic acids.

The UV-Vis absorption spectra of the four SiO₂-based nanomaterials (SiO₂, SiO₂/Bark, SiO₂/Res, SiO₂/Bark-Res) were recorded in the 200–700 nm range and are shown in Fig. 5. Each material exhibited a distinct electronic fingerprint, reflecting its unique organic-inorganic composition.

The spectra were recorded in solid state (powder form) using a quartz cell with a fixed path length. Absorbance (A) was measured directly; however, since the samples of the SiO₂-based nanomaterials were analyzed as powders, the absolute absorption coefficient α could not be calculated due to the lack of a uniformly defined film thickness and the

Table 1 Optical band gap parameters of SiO₂-based nanomaterials by Tauc plot analysis

Nanomaterial	Band gap E_g (eV)	Corresponding wavelength λ (nm)	Corresponding frequency ν ($\times 10^{15}$ Hz)	Corresponding spectral region
SiO ₂	5.5	225	1.33	Far UV (UV-C)
SiO ₂ /Bark	5.2	238	1.26	Far UV (UV-C)
SiO ₂ /Res	5.0	248	1.21	Far UV (UV-C)
SiO ₂ /Bark-Res	4.7	264	1.14	Far UV (UV-C)

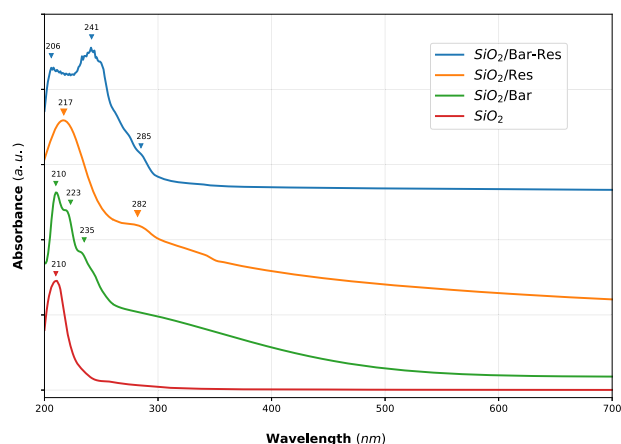


Fig. 5 UV-Vis spectra of SiO_2 -based nanomaterials. SiO_2 , SiO_2/Bar , SiO_2/Res , $\text{SiO}_2/\text{Bar-Res}$

presence of light scattering. Nevertheless, under constant measurement conditions, the absorbance remains proportional to the absorption by $\alpha h\nu = A(h\nu - E_g)^{1/n}$, allowing comparative analysis among materials. Tauc plots were constructed by plotting $(Ah\nu)^{1/n}$ versus photon energy ($h\nu$), assuming an indirect allowed transition ($n = 2$), a common approach for nanostructured materials [22, 23]. The band gap (E_g) was extracted by extrapolating the linear region of the Tauc plot to the photon energy axis [23]. Table 1 summarizes the optical band gap parameters of SiO_2 -based nanomaterials determined by the Tauc method.

SiO_2 displayed a single absorption band centered at 210 nm. This band is characteristic of oxygen-to-silicon charge transfer transitions ($\text{O}^{2-} \rightarrow \text{Si}^{4+}$) in the silica network [24]. Its Tauc plot yielded a band gap of 5.5 eV, consistent with values reported for silica [23].

SiO_2/Bar which incorporates pine bark extract, presented a more complex profile with absorption bands at 210 nm, 223 nm, and 235 nm. The band at 210 nm corresponds to the silica backbone, while the new bands at 223 and 235 nm are attributed to $\pi \rightarrow \pi^*$ transitions within phenolic moieties (e.g., tannins, flavonoids) and $n \rightarrow \pi^*$ transitions of carbonyl groups present in bark-derived polyphenols [25]. The progressive red shift of these bands suggest electronic interactions between the silica matrix and the organic phase. The band gap of SiO_2/Bar decreased to 5.2 eV, indicating a slight narrowing of the HOMO-LUMO gap due to the introduction of organic chromophores with lower energy transitions.

SiO_2/Res , formulated with pine resin, exhibited absorption bands at 217 nm and 282 nm. The band at 217 nm is assigned to isolated double bonds ($\text{C}=\text{C}$) and $\text{C}=\text{O}$ groups from resin acids and terpenoids, while the band at 282 nm corresponds to conjugated systems present in abietane-type resin acids and unsaturated terpenes [20, 26]. The presence of the 282 nm band confirms the incorporation of resin-

derived conjugated species into the nanomaterial. The band gap of SiO_2/Res was estimated as 5.0 eV.

$\text{SiO}_2/\text{Bark-Res}$ the most complex formulation combining both bark and resin components, showed absorption bands at 206 nm, 241 nm, and 285 nm. The band at 206 nm arises from the silica matrix. The bands at 241 nm and 285 nm reflect a broader distribution of electronic states, likely resulting from synergistic interactions between bark polyphenols and resin terpenoids within the silica network. This electronic coupling leads to the lowest band gap among all materials: 4.7 eV, as estimated by Tauc analysis.

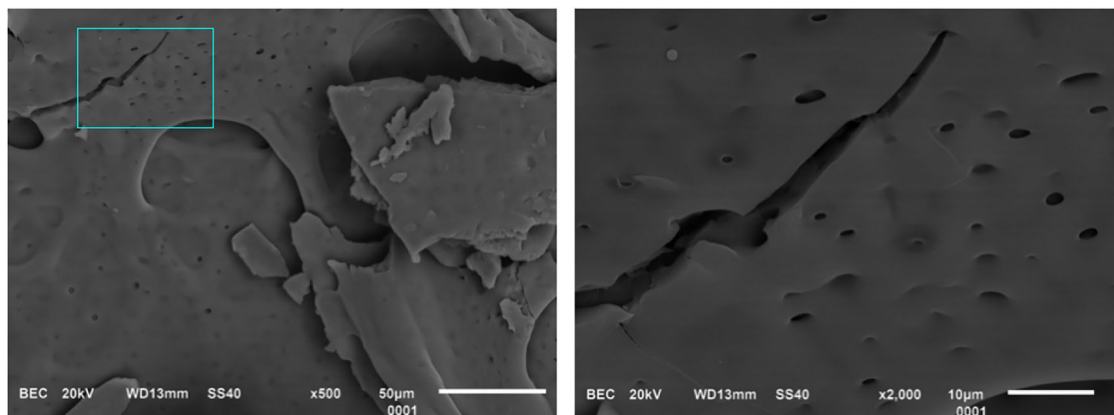
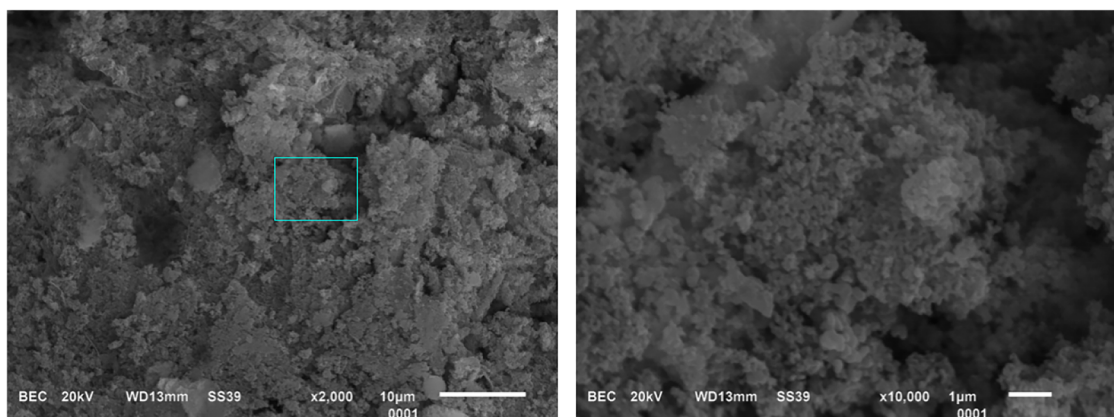
The progressive decrease in band gap from SiO_2 (5.5 eV) to $\text{SiO}_2/\text{Bark-Res}$ (4.7 eV) can be attributed to increasing electronic conjugation introduced by the organic components, rather than simply their overall structural complexity (mass and long structure). Interestingly, although SiO_2/Bark contains bark-derived compounds, where lignin accounts for 51.55% and holocellulose for 46 % and cellulose for 23.21 % [27] its band gap reduction (5.5 to 5.2 eV) is actually smaller than that of SiO_2/Res (5.5 to 5.0 eV), which incorporates pine resin composed of 70–80% α -pinene and related terpenes [26, 28]. This apparent paradox can be explained by the type and arrangement of conjugated systems present in each organic fraction. Despite the higher structural complexity and aromatic content of lignin, its extensive cross-linking and three-dimensional network may limit effective electronic coupling with the silica matrix. In contrast, the resin components particularly abietane-type resin acids with their conjugated diene systems, along with the high concentration of monoterpenes (α -pinene, Δ^3 -carene, camphene) appear to facilitate more efficient electronic delocalization at the organic-inorganic interface, despite their smaller molecular size and lower structural complexity. The synergistic combination of both bark and resin components in $\text{SiO}_2/\text{Bark-Res}$ yields the lowest band gap (4.7 eV), indicating that the two organic phases contribute additively or cooperatively to the electronic structure. This trend suggests enhanced electronic delocalization and possible formation of charge-transfer complexes at the organic-inorganic interfaces, which may influence not only the optical properties but also the reactivity and biological activity of the nanomaterial. Table 2 summarizes the main absorption bands, and the proposed electronic transitions for each SiO_2 -based nanomaterial.

Figure 6 show the Scanning electron microscopy (SEM) revealed that pure SiO_2 nanomaterials exhibit a plate-like morphology with oval-shaped cavities, resembling a porous "Gruyère cheese" texture. The average pore diameter is approximately 1.6×10^3 nm, (1600.0 nm), indicating an amorphous structure.

The micrography of SiO_2/Bar nanomaterial Fig. 7, also exhibits a granular morphology, with average particle clusters of 85 nm. The presence of polyphenols, phenolic

Table 2 UV-Vis absorption bands and Proposed electronic transitions of SiO₂-based nanomaterials

Nanomaterial	Absorption bands (nm)	Proposed electronic transitions
SiO ₂	210	O ²⁻ → Si ⁴⁺
SiO ₂ /Bark	210, 223, 235	$\pi \rightarrow \pi^*$ (phenolic/tannins), $n \rightarrow \pi^*$ (C=O)
SiO ₂ /Res	217, 282	Isolated C=C/C=O (217 nm); Conjugated systems (282 nm)
SiO ₂ /Bark-Res	206, 241, 285	Combined contributions from bark and resin components

**Fig. 6** Scanning electron micrographs of the SiO₂ nanomaterial at different magnifications (left: × 500; right: × 2000)**Fig. 7** Scanning Electron Microscopy (SEM) of SiO₂/Bar nanomaterial, left: × 2000; right: × 10,000

acids, and lignin in pine bark induces a chemical oxidation reaction that disrupts the lamellar texture of pure SiO₂, transforming it into fine, amorphous particle aggregates.

This transformation from large, plate-like structures to granular particle clusters signifies a shift in SiO₂'s physicochemical properties, reflecting true nanomaterial behavior with average particle sizes in the nanometer range.

Figure 8 shows the microscopy of SiO₂/Re nanocomposite shows a granular plate morphology with semi-hemispherical protrusions. Microstructural analysis indicates that the resin encapsulates the SiO₂, forming growth nuclei with an average diameter of 598 nm. These nuclei are uniformly distributed on resin plates featuring oval pores with a mean diameter of 1.3×10^3 nm (1300 nm).

Figure 9 shows the microscopy of SiO₂/Bar-Res nanomaterial, the combined presence of pine bark and resin leads to a mixed morphology that includes contact evolution nuclei similar to those seen in SiO₂/Bar. This combination of granular agglomerates and nucleation points within the SiO₂ matrix introduces charge density variations, increasing the number of active surface sites. As a result, the synergistic interaction between resin acids and bark-derived bioactive compounds enhances the material's insecticidal potential against *T. aniceps* found on avocado tree leaves.

EDS results (Table 3) show the presence of carbon confirmed organic incorporation in the SiO₂ matrix, aligns with the surface interactions of nanoformulations reported in recent studies [4]. The decrease in the mass percentage of

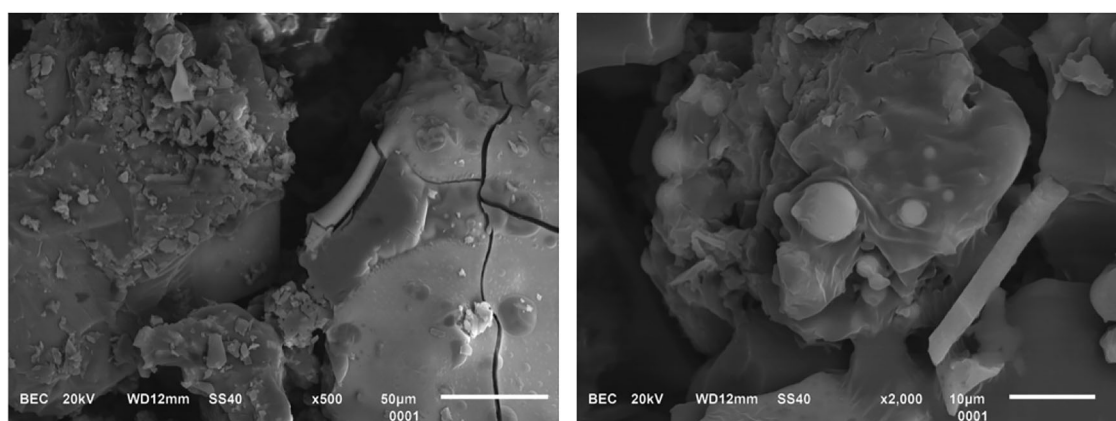


Fig. 8 Scanning electron micrographs of the SiO_2/Re nanomaterial at different regions, different magnifications (left: $\times 500$; right: $\times 2000$)

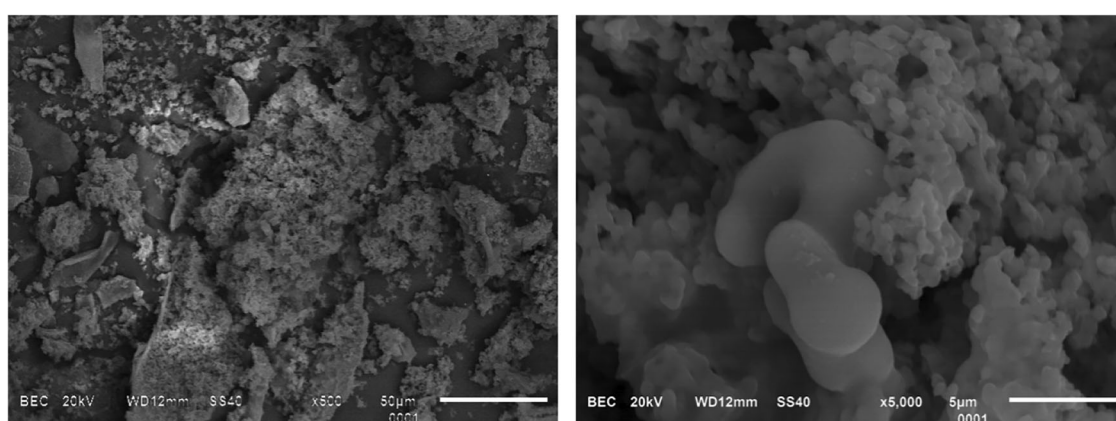


Fig. 9 Scanning Electron Microscopy (SEM) of $\text{SiO}_2/\text{Bar-Res}$ nanomaterial at different regions, different magnifications (left: $\times 500$; right: $\times 5000$)

Table 3 EDS info of based SiO_2 nano materials

	Silicon (Si) Wt. [%]	Oxygen (O) Wt. [%]	Carbon (C) Wt. [%]
SiO_2	45.19	54.81	0.0
SiO_2/Bar	26.62	45.07	28.31
SiO_2/Res	6.81	18.66	74.53
$\text{SiO}_2/\text{Bar-Res}$	20.21	45.14	34.65

oxygen in the SiO_2 nanomaterials that contain pine bark and resin, compared to pure SiO_2 nanomaterial, is attributed to the formation of a resin layer on the SiO_2 surface (SiO_2/Res nanomaterial); that is, the resin begins to cover specific areas of the SiO_2 nanomaterial surface. In the SiO_2/Bar nanomaterial, the active components of the pine bark interact with the active sites on the SiO_2 surface, thereby reducing the oxygen and silicon mass percentages. For the nanomaterial containing both pine bark and resin, the active compounds from the pine resin exhibit higher selectivity toward the active sites on the SiO_2 surface than those from

the bark; therefore, the resin does not block the selective sites reserved for the bark's active compounds. As a result, a decrease in oxygen mass percentage and a reduction in silicon content similar to the SiO_2/Bar nanomaterial is observed, as shown in Table 3 matching nanoformulation surface interactions [4].

Figure 10 presents the FTIR spectra of the SiO_2 -based nanomaterials. The analysis of the functional group region reveals vibrational modes associated with surface hydration and organic functionalization.

Vibration modes in the range of $3200 - 3900 \text{ cm}^{-1}$ and at $\sim 1640 \text{ cm}^{-1}$ are observed, corresponding to the O-H stretching (ν_{OH}) and H-O-H bending ($\delta_{\text{H-O-H}}$) modes of adsorbed water and surface silanols, respectively (purple zone). These features indicate a high degree of surface hydration in all materials [20, 21] following modern nanomaterial protocols [19]. Specifically, the broad and weak peaks at 3778 cm^{-1} (SiO_2), 3784 cm^{-1} (SiO_2/Bar), and 3776 cm^{-1} (SiO_2/Res) are assigned to free or isolated silanols. In contrast, the broader bands at 3385 cm^{-1} (SiO_2) and 3272 cm^{-1} (SiO_2/Res) are attributed to hydrogen-bonded hydroxyl groups.

The region between 3000 and 2800 cm^{-1} (gray zone) shows contributions from C-H stretching vibrations ($\nu_{\text{C-H}}$). In the pure SiO_2 sample, the bands at 2929 cm^{-1} (asymmetric CH_2 stretch) and $\sim 2929 \text{ cm}^{-1}$ (symmetric CH_2 stretch) are attributed to organic contamination, likely from residual ethoxy groups ($-\text{OCH}_2\text{CH}_3$) originating from the unreacted TEOS precursor used in the sol-gel synthesis [21, 29–32]. The presence of these bands is also confirmed in the functionalized nanomaterials (SiO_2/Bar , SiO_2/Res , $\text{SiO}_2/\text{Bar-Res}$).

The successful surface silanization is further evidenced by the distinct C-H stretching modes at 2961 cm^{-1} (SiO_2/Bar and SiO_2/Res) and 2931 cm^{-1} ($\text{SiO}_2/\text{Bar-Res}$), characteristic of the methyl and methylene groups introduced by the organosilane agents [21].

A weak contribution from carbonyl (C = O) groups is detected around 1700 cm^{-1} (yellow zone), with a band at 1690 cm^{-1} for *SiO₂/Res* and a minor feature for *SiO₂/Bar-Res*. In the fingerprint region (1500–400 cm^{-1}), the bands between 1465 and 1430 cm^{-1} (observed for all samples) can

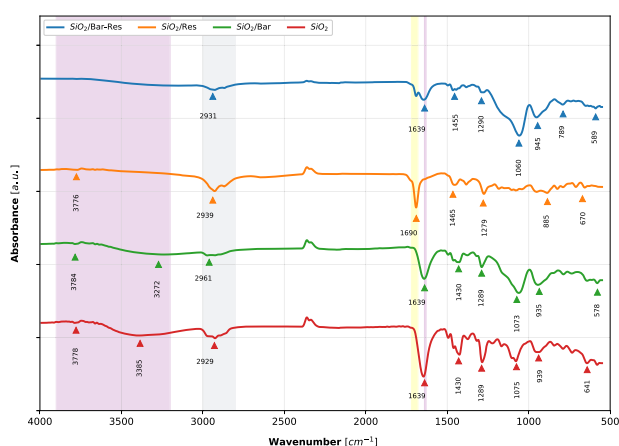


Fig. 10 FTIR spectra of SiO₂-based nanomaterials: SiO₂, SiO₂/Bar, SiO₂/Res, SiO₂/Bar-Res

be assigned to a combination of several vibrational modes, including bending vibrations of silicon ethoxide species ($\equiv \text{Si}-\text{OEt}$), C-H deformation modes (δ_{CH_3}), and symmetric carboxylate stretches (ν_{COO}) [20].

The most prominent feature is the asymmetric Si-O-Si stretching vibration, which appears as a composite band for the SiO₂, SiO₂/Bar, and SiO₂/Bar-Res samples. This band is characterized by a transverse optical (TO) mode component at $\sim 1075\text{--}1060\text{ cm}^{-1}$ and a longitudinal optical (LO) mode shoulder at $\sim 1290\text{--}1289\text{ cm}^{-1}$, a signature of the amorphous silica network obtained via the sol-gel method [33, 34]. The SiO₂/Res spectrum shows only the TO mode component at 1279 cm^{-1} .

Additional bands in the low-wavenumber region are assigned to the Si-O stretching mode of silanols at $\sim 939\text{ cm}^{-1}$ (SiO_2) and the O-Si-O rocking vibration at $\sim 641\text{ cm}^{-1}$ (SiO_2). For the functionalized materials, the bands at 935 and 578 cm^{-1} (SiO_2/Bar), 885 and 670 cm^{-1} (SiO_2/Res), and 945 , 789 , and 589 cm^{-1} ($\text{SiO}_2/\text{Bar-Res}$) are associated with Si-oxygen (Si-O) vibrational modes, confirming the incorporation of the respective Bark extract, Resin and the Bark and Resin extract [21].

Controlled hydrolysis of TEOS in acidic or basic media, using ethanol as cosolvent and water as hydrolyzing agent, yields partially condensed silicon oligomers featuring terminal --Si-OH groups (silanols, polar and bidirectional in H-bonds) and $\text{--Si-OCH}_2\text{CH}_3$ groups (residual ethoxys, susceptible to transesterification under heating), alongside inert bridging siloxanes (--Si-O-Si--).

Pine resin, composed of resin acids, terpenes, and terpenic alcohols, interacts as follows (Fig. 11a): interaction 1 shows resin acids esterified with ethanol at 70 °C reflux (R-COOEt) forming weak $C=O \cdots HO-Si$ hydrogen bonds and hydrocarbon van der Waals forces over silanols; interaction 2 involves terpenic alcohols reacting with ethoxy groups via transesterification, generating the most permanent covalent bond despite their low relative concentration.

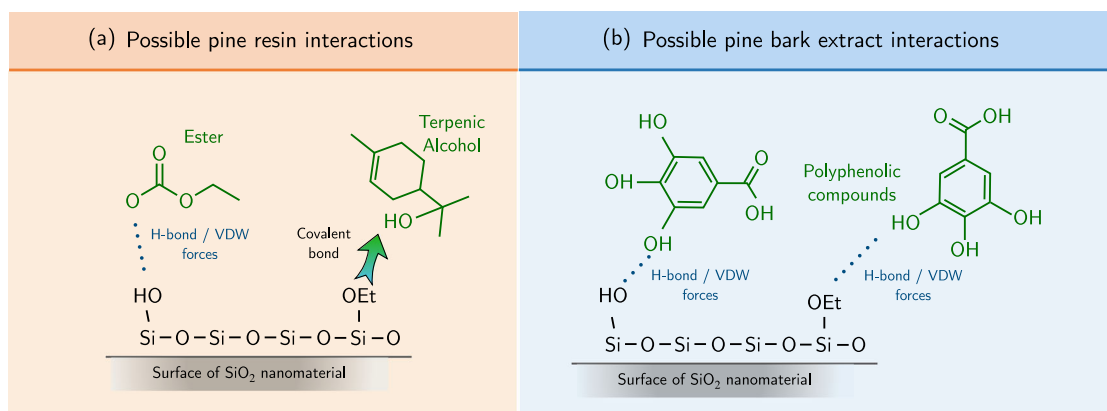


Fig. 11 Proposed representation of the interactions between silicon oxide oligomers (surface of an SiO₂ nanomaterial) and the different families of compounds present in natural extracts: **a** terpene esters and alcohols from pine resin, and **b** polyphenols from pine bark

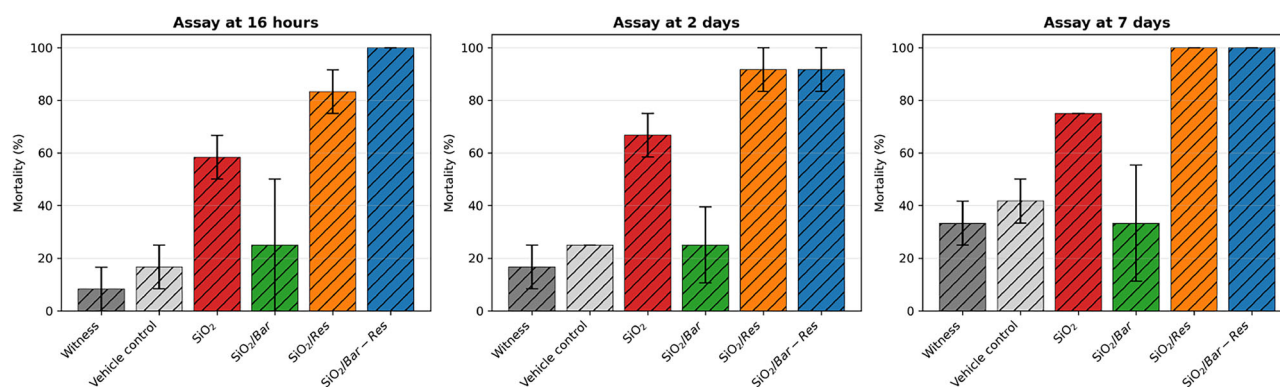


Fig. 12 Mortality by treatment at each time point

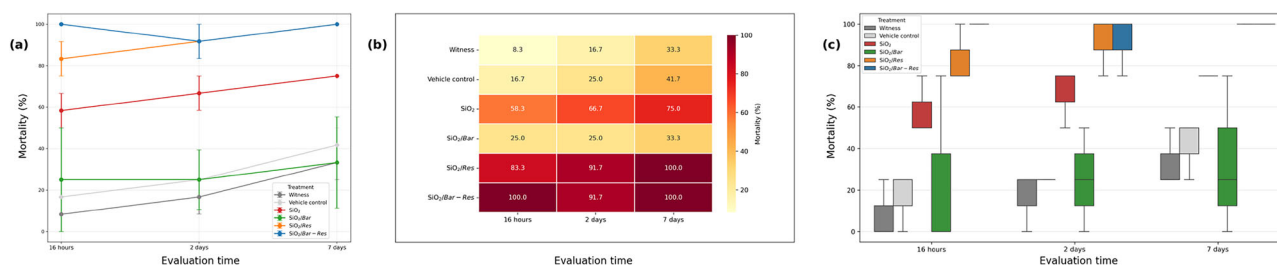


Fig. 13 **a** Temporal evolution of nymph mortality, **b** Heatmap of treatment along the time, **c** distribution of mortality

The extract of pine bark is a complex mixture of polyphenolic compounds, polysaccharide fragments, and lignin derivatives. *Pinus pinaster* bark contains lignin and polyphenols (ca. 44%), polysaccharides (ca. 39%), and extractives (ca. 17%). After alkaline extraction, the soluble fraction is enriched in condensed tannins, along with lignin fragments (Mn 1000–1500 Da) and hemicellulosic sugars [35]. This motivate that the polyphenolic components both tannins and lignin (derived phenols) interact with the surface of the silicon oxide oligomer. The polyphenolic compounds, which bear multiple –OH groups, interact with the oligomer surface through multiple hydrogen bonds between their phenolic –OH groups and the surface silanols. These bonds are complemented by London dispersion forces between the aromatic rings and the hydrophobic regions of the oligomer surface (including ethoxy groups and structural defects), as shown in Fig. 11b. The surface heterogeneity of the oligomers favors the preferential adsorption of polyphenols onto the higher-energy surface regions.

3.2 Laboratory evaluation of nanomaterial induced bio-inhibition on *Trioza anceps*

The mortality of *Trioza anceps* nymphs at each evaluation time (16 h, 48 h, and 7 days) is presented in Fig. 12. The SiO₂/Bar-Res nanocomposite achieved 100% nymphal mortality, demonstrating superior efficacy comparable to recent nanopesticide developments [2]. Each bar plot

corresponds to an independent and destructive experiment in which six treatments were evaluated with three replicates each. The bars represent the mean mortality percentage per treatment at each time point, while the error bars indicate the standard error of the mean (SEM). This metric estimates the precision of the sample mean and its potential variability in the population, making it more suitable than the standard deviation (SD) -which describes the dispersion of individual measurements- for comparing means across treatments and time points.

The six evaluated treatments comprised a negative control called Witness (distilled water), a Vehicle control (ethyl alcohol), and four nanomaterials designated as SiO₂, SiO₂/Bar, SiO₂/Res, and SiO₂/Bar-Res. The inclusion of a vehicle control was essential to discard any potential effect of the solvent on nymph mortality, as all nanomaterials were diluted in a 30:1 ratio.

The temporal evolution of nymph mortality Fig. 13a was constructed using mean values derived from the experimental data. The arithmetic mean of the three replicates was then calculated to represent the central tendency of mortality for each treatment at each time point. To provide a measure of the precision of these estimated means, the standard error of the mean (SEM) was computed as the standard deviation of the three replicates divided by the square root of the sample size ($\sqrt{3}$). The resulting mean values were plotted over time, with error bars representing the SEM, allowing for a visual assessment of treatment

performance and variability prior to inferential statistical analysis. The resulting plot shows that three treatments exhibited little to no effectiveness and remain clearly separated from the others: the Witness, the Vehicle control, and SiO_2/Bar . Notably, no increasing trend was observed in these treatments over time, as mortality remained consistently low. This flat trajectory supports the conclusion that mortality is not attributable to the solvent (ethanol) but rather to the nanomaterials themselves.

Another important observation concerns SiO_2/Bar , whose formulation includes components analogous to those found in bark. This is particularly relevant given that bark's natural function in a tree trunk is to act as a barrier between the internal tissues and the external environment. In this context, the lack of a temporal increase in SiO_2/Bar -induced mortality may reflect a protective mechanism, which would explain the low nymph mortality.

Regarding SiO_2 , the base nanomaterial demonstrated a clear capacity to induce nymph mortality. However, its moderate efficacy, compared to the other SiO_2 -derived formulations, suggests that while the base nanomaterial is conceptually sound, its insecticidal activity may be enhanced through structural modifications or combination with other components. In contrast, the temporal evolution of SiO_2/Res , and $\text{SiO}_2/\text{Bar-Res}$ revealed a markedly higher efficacy, achieving 100% mortality by day 7 in all replicates.

It is also worth noting that all SiO_2 -based treatments are insoluble in water. This physicochemical property has important implications for field applications: under natural conditions, rainfall is unlikely to wash off the formulations, potentially extending their residual activity beyond the 7 day window observed in this laboratory bioassay.

To explore the overall patterns of treatment efficacy across time, a heatmap was constructed (Fig. 13b). Each cell represents the mean mortality percentage for a given treatment at a specific evaluation time, with a color gradient ranging from light for low mortality to dark for high mortality. This visualization allows for a rapid and intuitive comparison of treatment performance, revealing that Witness, Vehicle control, and SiO_2/Bar consistently exhibited low mortality across all time points, while SiO_2/Res and $\text{SiO}_2/\text{Bar-Res}$ showed progressively darker cells, reaching the highest mortality by day 7. The heatmap thus provides an initial visual overview of the data, highlighting candidate treatments for further evaluation.

While the temporal evolution (Fig. 13a) and heatmap (Fig. 13b) provide an overview of treatment efficacy based on mean values, the distribution plot (Fig. 13c) offers critical insight into the consistency and reliability of each treatment. This visualization reveals not only the central tendency but also the dispersion of individual replicates,

allowing for the identification of treatments with high variability versus those with stable, reproducible effects.

Notably, treatments such as SiO_2/Res and $\text{SiO}_2/\text{Bar-Res}$ exhibited low variability, with all replicates clustered near 100% mortality by day 7, indicating high consistency and predictability. In contrast, SiO_2/Bar showed wide dispersion across all time points, with individual replicates ranging from 0% to 75% mortality, reflecting erratic performance and limited reliability. The control and vehicle control treatments maintained consistently low mortality with minimal dispersion, reinforcing their role as baselines.

This analysis highlights that effectiveness alone is insufficient; a treatment must also demonstrate reproducibility to be considered viable for field applications. The distribution plot thus serves as a crucial tool for quality evaluation of treatment performance beyond mean based comparisons.

On the other hand, given that the response variable is the count of dead nymphs out of a fixed total, four galls per leaf, the data follow a binomial distribution rather than a continuous normal distribution. Therefore, a Generalized Linear Model (GLM) with binomial family and logit link function was fitted to the mortality data, instead of a traditional ANOVA, which assumes normality and homoscedasticity. This approach is statistically appropriate for proportion data and avoids biases associated with arcsine transformations or analyses based on percentages.

The GLM revealed significant differences in nymph mortality between treatments, with $\chi^2 = 107.16$, $\text{df} = 5$, $p < 0.0001$, indicating a highly significant effect between formulations. The extremely low p -value means that the probability of observing such differences by chance is less than 0.01%. This suggests that the treatments exhibit markedly different levels of efficacy, allowing for the selection of a more promising candidate for field validation.

Furthermore, the GLM analysis showed a barely significant effect of exposure time with $\chi^2 = 6.14$, $\text{df} = 2$, $p = 0.0465$, indicating that nymph mortality with respect to time is right at the limit of statistical significance ($p < 0.05$). Mortality changes over time, but the effect is not as significant as the difference between treatments. It can be said that the contact time with the nanomaterial influences insecticidal efficacy; this can be tested with field trials and by measuring the degradation time of the dissolved nanomaterial after application to gall-affected leaves.

Likewise, the interaction between treatment and time was not significant ($\chi^2 = 4.52$, $\text{df} = 10$, $p = 0.9206$), indicating that the relative performance of the treatments remained consistent over time. That is, the most effective treatments at 16 h remained so at 7 days, and the least effective treatments maintained their poor performance throughout the experiment.

Table 4 Dunnett's test comparing each nanomaterial against the Vehicle control at 7 days post-treatment

Comparison	Difference (%)	Adjusted <i>p</i> -value	Significance
SiO ₂ /Res vs. Vehicle control	58.3	0.016	*
SiO ₂ /Bar-Res vs. Vehicle control	50.0	0.016	*
SiO ₂ vs. Vehicle control	33.3	0.262	ns
SiO ₂ /Bar vs. Vehicle control	-8.3	0.991	ns
Witness vs. Vehicle control	-8.3	0.991	ns

* = significant difference ($p < 0.05$); ns = not significant ($p > 0.05$)

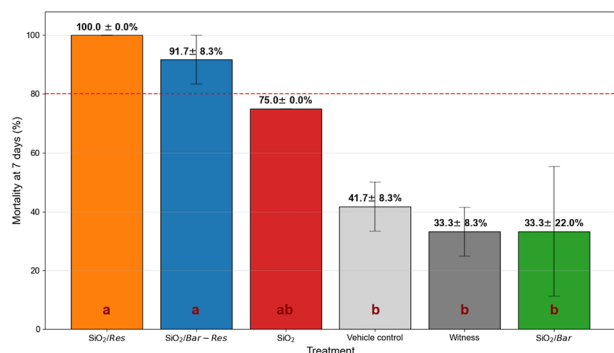


Fig. 14 Mortality of *Trioza* aneeps nymphs at 7 days post-treatment. Bars represent mean mortality (%) $\pm$ SEM ($n = 3$). Different lowercase letters in bars indicate significant differences among treatments according to Tukey's HSD test ($\alpha = 0.05$). The dashed red line marks the 80% mortality threshold considered for candidate selection. SiO₂/Res and SiO₂/Bar-Res were selected as promising candidates for field validation

To determine which treatments outperformed the formulation base, Dunnett's test was performed (Table 4) using the Vehicle control as the reference treatment, as it represents the formulation base without the active nanomaterials. The results showed that SiO₂/Res and SiO₂/Bar-Res induced significantly higher mortality than the Vehicle control ($p = 0.016$), with differences of 58.3% and 50.0%, respectively. In contrast, SiO₂, SiO₂/Bar, and Witness (water) did not differ significantly from the Vehicle control ($p > 0.05$). Notably, the lack of significant difference between the Witness and Vehicle control treatments ($p > 0.05$) confirms that the solvent (ethanol) itself has no insecticidal effect, and therefore any mortality observed in the nanomaterial treatments can be attributed to the active compounds rather than the formulation base.

Regarding the post-hoc test, based on the mean mortality values derived from three independent replicates ($n = 3$) at 7 days post-treatment, the highest mortality was achieved by SiO₂/Res and SiO₂/Bar-Res, were the only treatments with mortality significantly higher, 100% and 91.7% mortality respectively. SiO₂ showed intermediate efficacy with 75.0% mortality (SEM = 0.0%). In contrast, Vehicle control (41.7 $\pm$ 8.3%), Witness (33.3 $\pm$ 8.3%), and SiO₂/Bar (33.3 $\pm$ 22.0%) exhibited considerably lower mortality rates (Fig. 14).

Tukey's HSD post-hoc test ($\alpha = 0.05$) was used to compare all treatments among themselves, revealing the following grouping:

- **Group a (highest efficacy):** SiO₂/Res and SiO₂/Bar-Res (100%), significantly different from Witness ($p = 0.006$), Vehicle control ($p = 0.0158$), and SiO₂/Bar ($p = 0.006$).
- **Group ab (intermediate):** SiO₂ (75%), which did not differ significantly from other groups ($p > 0.05$).
- **Group b (lowest efficacy):** Vehicle control (41.7%), SiO₂/Bar (33.3%), and Witness (33.3%).

In summary, SiO₂/Res and SiO₂/Bar-Res were selected as promising candidates for field validation, as they consistently exceeded the 80% mortality threshold and demonstrated statistically significant efficacy compared to the control treatments. Although both SiO₂/Res and SiO₂/Bar-Res exhibited high mortality rates and belonged to the same statistically homogeneous group (Tukey's HSD), SiO₂/Res offers a significant practical advantage, it consists of a single compound, making it easier to synthesize, formulate, and scale up compared to more complex formulations. Its simpler composition reduces production costs and potential variability, while maintaining maximum efficacy. Therefore, SiO₂/Res was prioritized over SiO₂/Bar-Res for subsequent field trials, where its performance under natural conditions will be evaluated.

3.3 Field validation

The field study was conducted in four avocado orchards of different sizes. In two orchards with over 50 trees each, 12 trees per orchard were selected for treatment. In a third orchard with 20 trees, 7 trees were selected, and in the smallest orchard with 12 trees, 4 trees were chosen. All selected trees were tagged with colored ribbons (Fig. 15) and sprayed with SiO₂/Res at a higher dilution than in laboratory assays, with applications at day 1 and day 15. The dilution was adjusted based on the observed number of galls per tree and the limited availability of the nanomaterial, which was insufficient to cover the full of all selected trees at the laboratory concentration.

However, most trees were lost when farmers, mistaking the whitish layer of the nanomaterial for a fungal pathogen,

Fig. 15 Left: Visual identification of experimental trees (Field conditions) using colored ribbons. **Right:** Foliar application of SiO_2/Res treatment under field conditions



Fig. 16 a Initial state, leaf with a greenish-yellow gall shortly after SiO_2/Res application. The whitish layer of the nanomaterial is visible on the leaf surface. **b** Same leaf four months after treatment, the gall has turned brown and dried out, indicating complete nymph mortality. The whitish residue of SiO_2/Res remains visible on the leaf surface



pruned and burned the branches, forcing the premature termination of the field trial by the third visit (day 45). This drastic measure, combined with the removal of identification ribbons, left only four trees from a single orchard available for continued observation, those whose owners trusted both the nanomaterial and our team.

This experience underscores a critical challenge in field research, community perception and trust can be as decisive as experimental design, particularly when visible residues resemble unfamiliar or suspicious agents. Despite these limitations, the surviving trees provided valuable preliminary insights into the performance of SiO_2/Res under real conditions.

3.4 Field observation and implications

The persistence of SiO_2/Res for up to four months (Fig. 16), even after heavy rainfall, highlights its potential for long-term control of *Trioza anceps* Tuthill. This rainfastness can be attributed to the intrinsic hydrophobic nature of the synthesized nanomaterial, further enhanced by the presence of pine resin (*Res*), which contributes to the formation of a protective layer against abiotic and biotic factors. Similar hydrophobic nature has been reported for silica-based formulations to which a hydrocarbon or hydroxyl groups has been added [36, 37].

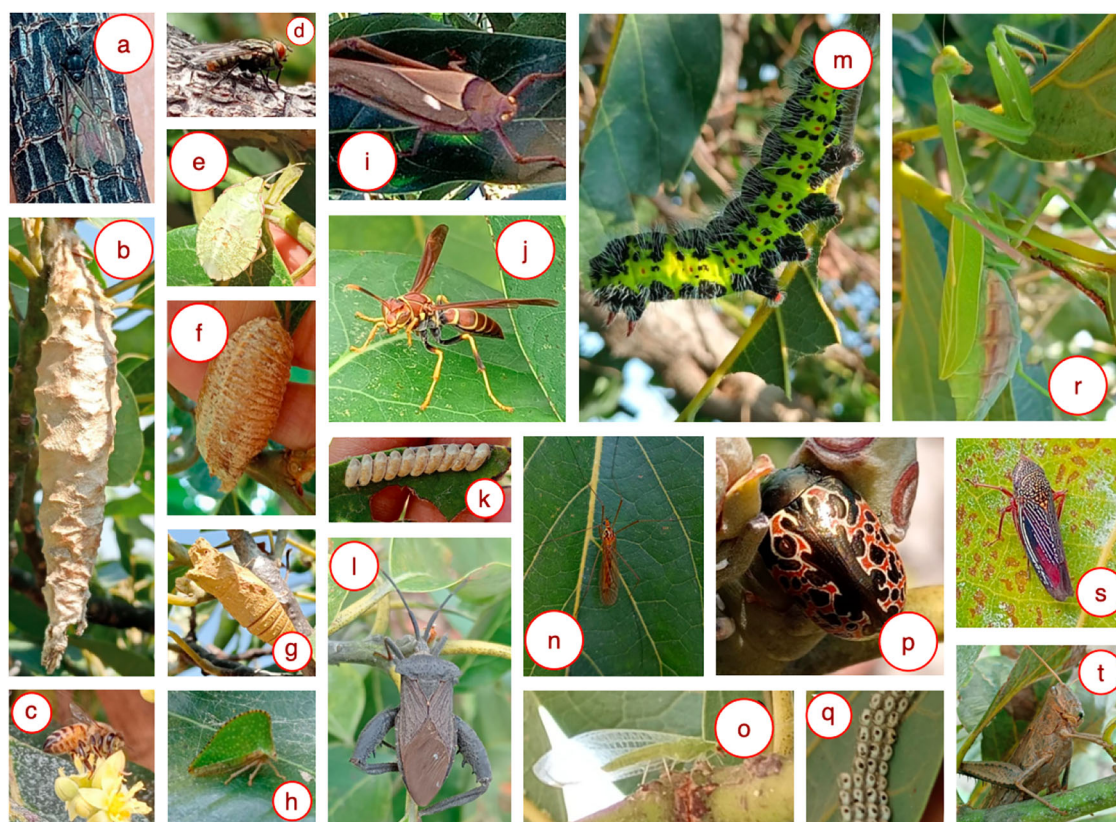


Fig. 17 Non-target entomofauna observed after application of the specific insecticide against *Trioza anceps*. **a** Parasitoid wasp (Hymenoptera), **b** Psychidae, **c** *Apis mellifera*, pollinating, **d** Dipterous, **e** Hemipteran, **f** Ootheca, **g** Lepidopteran pupa, **h** Stictoccephala, **i** Orthopter, **j** Hymenopter, **k** Orthopteran eggs, **l** Hemipteran, **m** Lepidopteran larva, **n** Crane-fly, **o** Lacewing, **p** Coleoptera: Chrysomelidae, **q** Hemipteran eggs, **r** Mantodea, **s** Cicadellidae, **t** Acrididae

The lack of visible damage to non-target entomofauna, including predators and pollinators, supports the biocompatibility of SiO_2 nanomaterials [38]. Although some authors have reported morphological alterations in chewing insects exposed to silicates [18, 39], these effects were not evaluated here, as application occurred after oviposition and was not aimed at direct contact with adult insects.

Regarding the surrounding entomofauna (Fig. 17) and the inhibition of *Trioza anceps* nymphs, the treatment SiO_2/Res exhibited high specificity, effectively controlling the target pest while preserving the stability of non-target insect populations. A healthy mix of functional guilds was maintained, including herbivores (various species of true bugs, caterpillars, and grasshoppers) and, most importantly, beneficial organisms such as predators (mantises, lacewings) and pollinators (bees). Furthermore, the confirmed presence of multiple life stages (from eggs and oothecae to pupae and adults) indicates that the insecticide did not interfere with the developmental or reproductive processes of the local fauna.

This selectivity suggests that M3 could be applied only to visibly damaged leaves or localized infestations, rather than as a full-coverage spray. Such a targeted approach would further minimize exposure to non-target organisms, potentially enhancing the conservation of beneficial

entomofauna while maintaining effective pest control. This strategy aligns with integrated pest management (IPM) principles, where chemical interventions are used only when and where necessary.

Four months after application, an observed improvement in fruit growth was noted in treated trees. This may be related to the induction of systemic acquired resistance (SAR) triggered by silicon polymerization in cell walls. This process promotes the accumulation of lignin and phenolic compounds, increases the activity of defense-related enzymes such as chitinases and peroxidases, and reduces the nutritional quality and digestibility of plant tissues for herbivores [40–42]. However, given that silicon transport in plants is limited and deposits tend to remain near the site of application [43], future studies should explore root uptake or soil application to enhance systemic effects and maximize the potential benefits of silicon-based nanomaterials.

4 Conclusion

This study evaluated the induced bio-inhibition of four SiO_2 -based nanomaterials (SiO_2/Bar , SiO_2/Res and $\text{SiO}_2/\text{Bar-Res}$) against *Trioza anceps* nymphs under controlled

laboratory conditions, and provided preliminary field observations on the most promising candidate, *SiO₂/Res*.

Characterization studies of the synthesized nanomaterials provide crucial insights into the structure underlying their insecticidal activity. UV-Vis spectrophotometry revealed distinct optical fingerprints and band gap energies (5.5 to 4.7 eV), correlating with the increasing electronic conjugation introduced by the organic components. Scanning electron microscopy (SEM) elucidated the morphology and the size of nanostructures in *SiO₂*-based nanomaterials. Fourier transform infrared spectroscopy (FTIR) identified surface functional groups (Si-O-Si, O-H, C-H, C=O) responsible for interactions between the organic components added to the *SiO₂* matrix, while also providing evidence for the loading and potential release of active components. Together, these characterization data are essential for proposing a molecular model of the chemical bonding between the representative organic compounds (bark tannins and resin terpenoids) and the *SiO₂* matrix, involving covalent Si-OR bonds and hydrogen bonding at the organic-inorganic interface.

The laboratory bioassays demonstrated that *SiO₂/Res* and *SiO₂/Bar-Res* were the most effective treatments, achieving 100% and 91.7% mortality at 7 days post-application, respectively. Both treatments significantly outperformed the Witness (only water) and Vehicle controls ($p < 0.05$), and were statistically grouped in the highest efficacy category (Tukey's HSD, group a). *SiO₂* showed moderate efficacy (75%, group ab), while *SiO₂/Bar* did not differ from the controls (group b). The Generalized Linear Model confirmed significant effects of treatment ($\chi^2 = 107.16$, $p < 0.0001$) and time ($\chi^2 = 6.14$, $p = 0.0465$), with no interaction, indicating consistent treatment performance across time points.

Based on these results, *SiO₂/Res* was selected as the primary candidate for field validation due to its 100% mortality, zero variability (SEM = 0.0%), and simpler formulation compared to *SiO₂/Bar-Res*.

Field observations, although limited by farmer intervention, revealed that *SiO₂/Res* remained visible on leaves for up to four months after application, even after heavy rainfall, indicating strong rainfastness. Importantly, no adverse effects were observed on non-target entomofauna, including predators (mantises, lacewings), pollinators (bees), and parasitoids, supporting the biocompatibility of *SiO₂* nanomaterials. Additionally, treated trees showed enhanced fruit growth four months after application, possibly linked to the induction of systemic acquired resistance (SAR) via silicon polymerization in cell walls.

The premature termination of the field trial due to farmer intervention highlights a critical challenge in applied research: community perception and trust are as essential as experimental design. The whitish residue of *SiO₂/Res* was

mistaken for a fungal pathogen, leading to the pruning and burning of treated branches, this experience underscores the need for participatory approaches and clear communication with local stakeholders in future studies. In conclusion, *SiO₂/Res* is a promising candidate for integrated pest management of *Trioza* aneeps, combining high efficacy, environmental safety, and potential biostimulant effects. Future research should focus on:

- Field validation with larger sample sizes and multiple orchards,
- Optimization of application protocols, including concentration and timing,
- Exploration of root uptake or soil application to enhance systemic effects,
- Long-term studies on fruit yield and quality, and
- Community engagement strategies to facilitate technology adoption.

Data availability

No datasets were generated or analysed during the current study.

Acknowledgements This work was supported by Benemérita Universidad Autónoma de Puebla.

Author contributions J.A.M.R. and A.A.G. conceived the study. J.A.M.R. developed the methodology and designed the experiments. J.A.M.R. and E.R.R. conducted the investigation and formal analysis. G.C.A. and E.A.V.T. curated the data and prepared the visualizations. G. C. A. wrote the original draft. E.A.V.T. and L.A.M.R. reviewed and edited the manuscript. J. A. M. R. and L.A.M.R. supervised the project. All authors have read and agreed to the published version of the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

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